

NEUQUEN, Argentina

WMO: 877150

Lat: 38.952S

Lon: 68.137W

Elev: 889

StdP: 14.23

Time Zone: -3.00 (W03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	25.3	28.2	12.6	10.8	56.0	16.2	12.9	53.3	27.8	58.6	23.1	57.1	2.3	270	0.474

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	26.4	95.4	66.0	92.7	64.7	89.7	63.4	70.5	87.5	68.2	85.2	66.3	83.5	8.4	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.7	94.5	78.1	62.4	87.1	75.8	59.9	79.6	73.5	35.0	87.4	33.1	85.4	31.4	83.4	84.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 24.5	20.2	17.2	DB	20.8	101.5	3.1	2.3	18.5	103.2	16.7	104.5	14.9	105.8	12.7	107.5	(4)
(5)			WB	19.7	74.4	3.3	3.4	17.3	76.9	15.3	78.9	13.5	80.8	11.0	83.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	59.4	75.1	72.7	66.7	58.0	50.3	45.3	44.3	48.2	53.5	60.1	66.8	72.4		
	DBStd	12.18	5.45	6.00	6.09	6.13	5.83	5.70	5.58	6.07	5.80	5.95	5.76	5.77		
	HDD50	565	0	0	0	7	66	161	192	107	29	3	0	0		
	HDD65	3067	4	8	52	222	455	590	642	520	346	173	46	9		
	CDD50	3996	779	636	517	247	77	21	15	52	135	317	505	695		
	CDD65	1023	317	225	104	12	1	0	0	2	22	100	240			
	CDH74	11972	3679	2534	1210	196	13	1	0	17	59	339	1155	2769		
	CDH80	5038	1771	1143	429	34	1	0	0	2	9	74	367	1208		
(14) Wind	WSAvg	6.6	7.9	7.0	6.0	5.4	4.8	5.4	5.3	6.1	6.9	7.6	8.2	8.6		
(15) Precipitation	PrecAvg	8.6	0.6	0.6	0.7	1.0	1.0	0.7	0.6	0.4	0.6	1.4	0.6	0.4		
	PrecMax	18.2	2.1	3.5	4.4	9.1	3.4	2.4	1.5	1.9	2.4	7.8	3.5	2.2		
	PrecMin	3.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
	PrecStd	4.3	0.6	0.8	0.9	1.7	1.1	0.7	0.5	0.5	0.6	1.7	0.9	0.6		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	100.4	98.3	93.8	84.0	75.3	69.0	68.2	75.9	80.3	86.4	92.9	98.0		
		MCWB	68.4	69.3	65.1	61.8	56.4	52.1	51.5	54.2	56.7	60.3	63.0	66.4		
	2%	DB	95.4	94.1	88.8	79.4	69.5	63.6	62.9	69.3	74.6	81.0	87.9	93.6		
		MCWB	65.6	66.8	64.2	59.5	54.2	49.7	48.7	51.8	54.0	57.2	61.3	64.4		
	5%	DB	92.7	91.0	85.0	75.2	65.1	60.1	59.3	64.5	70.6	77.4	84.2	90.0		
		MCWB	64.5	65.4	63.4	57.5	52.4	48.0	47.1	49.3	51.8	55.5	59.5	62.6		
	10%	DB	89.4	87.4	80.8	71.4	62.2	56.9	55.9	60.6	67.2	73.7	80.7	87.2		
		MCWB	62.8	64.0	61.4	55.9	51.1	46.7	45.4	47.2	50.2	53.9	57.7	61.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	73.0	74.8	71.1	64.3	59.4	54.1	53.2	56.6	59.6	63.2	67.4	71.6		
		MCDB	91.9	90.8	85.3	78.1	67.7	64.7	63.8	71.9	74.1	80.8	86.1	90.4		
	2%	WB	69.5	70.9	67.8	61.7	56.1	51.2	50.3	52.6	55.7	60.1	63.6	67.8		
		MCDB	88.2	86.9	81.9	73.8	65.4	60.5	60.4	65.7	70.7	75.4	82.4	86.7		
	5%	WB	67.2	68.3	65.4	59.6	54.1	49.3	48.4	50.3	53.5	57.7	61.3	65.4		
		MCDB	85.6	84.8	79.6	70.8	63.2	57.4	57.7	62.0	67.0	72.9	79.6	84.4		
	10%	WB	65.1	66.2	63.2	57.6	52.3	47.6	46.2	48.5	51.4	55.6	59.2	63.1		
		MCDB	84.0	82.9	77.6	68.4	60.0	55.0	54.4	58.8	64.4	70.1	77.6	82.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	26.4	26.0	25.5	24.1	20.4	19.7	21.2	22.8	23.6	23.5	24.9	25.7		
		MCDBR	31.1	31.2	31.9	30.3	26.0	24.1	26.6	28.9	30.2	29.6	29.8	30.2		
		MCWBR	11.7	12.0	13.1	14.1	14.3	13.6	15.5	15.2	14.9	12.6	12.1	11.8		
	5% WB	MCDBR	28.4	27.6	27.3	26.1	22.9	21.3	24.5	26.3	26.9	26.9	27.4	28.4		
		MCWBR	12.8	12.7	13.1	13.6	13.7	13.3	15.2	14.7	14.3	12.8	12.7	13.0		
(40) Clear-Sky Solar Irradiance	taub		0.382	0.375	0.355	0.347	0.326	0.305	0.304	0.320	0.350	0.354	0.363	0.374		
	taud		2.321	2.353	2.415	2.418	2.469	2.540	2.536	2.477	2.375	2.367	2.352	2.342		
	Ebn at Noon		301	295	287	268	254	252	261	274	284	298	305	305		
	Edh at Noon		43	39	34	30	24	21	22	27	35	38	41	42		
(44) All-Sky Solar Radiation	RadAvg		2679	2292	1787	1205	745	608	689	970	1443	1922	2453	2720		
	RadStd		108	89	89	105	88	66	70	83	103	116	103	108		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	+0.50	N/A	N/A	N/A	+1.01	N/A	N/A	N/A	N/A	N/A	N/A
Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page